

FREDRICK MANDELA

SENIOR DATA & FULL-STACK ENGINEER | DATA PLATFORMS | BACKEND SYSTEMS

Nairobi, Kenya | +254 796 279 572 | fredricardomandela@gmail.com

[linkedin.com/in/fmandela](https://www.linkedin.com/in/fmandela) | github.com/fmandela | fredrickmandela.netlify.app

PROFESSIONAL SUMMARY

Senior data and full-stack engineer with 8+ years of experience building reliable data platforms, backend services, analytics products, integrations, and production applications. Combines hands-on engineering in Python, SQL, APIs, workflow orchestration, data quality, and cloud deployment with technical leadership across multi-country operations. Strong at translating ambiguous operational needs into observable, maintainable systems that teams can trust.

CORE COMPETENCIES

Data Platform Architecture | Backend & API Engineering | Analytics Engineering | Full-Stack Delivery | Workflow Orchestration | Data Quality & Observability | Financial Data Engineering | AI-Assisted Extraction | Engineering Leadership | Cloud & Container Deployment

PROFESSIONAL EXPERIENCE

ONE ACRE FUND, Nairobi, Kenya

Data Engineering Manager / Lead Data Engineer | Jan 2021 - Mar 2026

- Led three data engineers and a business analyst delivering production data platforms, integrations, and internal data products across 10+ country operations.
- Owned engineering prioritization across product, operations, finance, and analytics needs; reviewed designs, removed delivery blockers, and coordinated production releases.
- Built API- and database-driven integrations connecting operational systems, payment platforms, field applications, and analytics environments, reducing manual data movement.
- Built and operated infrastructure supporting 500M+ records and 5TB+ across Snowflake, SQL Server, Dataiku, Airflow, dbt, and Superset.
- Established automated quality checks, monitoring, alerts, incident follow-up, and ownership practices that improved production reliability and visibility.
- Led large-scale migration and synchronization programs and created reusable onboarding, validation, and rollout patterns for new country operations.
- Implemented governance patterns including role-based access, row-level security, validation, documentation, and trusted reporting layers for business users.

ONE ACRE FUND, Nairobi, Kenya

Data Engineer - IT Engineering | Jan 2018 - Dec 2020

- Designed ETL/ELT pipelines ingesting APIs, relational databases, operational systems, and third-party platforms into centralized warehouse environments.
- Developed integrations with Python, SQL, C#, REST APIs, and workflow automation tools to improve data availability across business systems.
- Helped replace legacy SSIS workflows with scalable automated pipelines, improving maintainability, monitoring, and refresh reliability.
- Partnered with BI, product, and business teams to define requirements, validate outputs, and deliver accurate datasets for operational and strategic decisions.

SELECTED PROJECTS

INVESTAR COMPANION - FINANCIAL DATA PLATFORM | Python, Playwright, OpenAI API, PostgreSQL/Supabase, Pydantic, GitHub Actions

- Built a platform that collects Kenyan unit-trust rates from provider websites, HTML, CSV, XLSX, and text-based PDFs, then normalizes them into auditable, analysis-ready records.
- Designed hybrid ingestion that uses deterministic provider parsers for stable sources and structured AI extraction for changing documents and layouts.
- Preserved source evidence, confidence, content hashes, canonical fund identity, effective dates, freshness, and extraction metadata for reliable comparison and audit.
- Created an API-ready market snapshot with rate movement, ranking, 30-day metrics, provenance, and publishing adapters for email, Slack, Telegram, and WhatsApp.

AI SQL ANALYST | Python, Streamlit, SQL, Applied AI

- Built a local-first analytics assistant that turns plain-English questions into validated SQL, executes queries against sample data, and explains results for non-technical users.

AI CAREER ASSISTANT | Python, Streamlit, Applied AI

- Built an assistant for creating professional resumes and assessing candidate-role fit from job descriptions, with an extension path to external job platforms.

DAIRY FARM RECORDS | Flutter, FastAPI, Python, REST APIs

- Developed a mobile and backend product for capturing animal, milk production, feeding, health, heat-cycle, and insemination records through authenticated APIs.

TECHNICAL PROFICIENCIES

Programming & backend: Python, SQL, C#, .NET, Node.js, FastAPI, REST APIs, webhooks, PostgreSQL, SQL Server

Data platforms & analytics: Snowflake, Airflow, dbt, Dataiku, Superset, Power BI, Airbyte, Kafka, RabbitMQ

AI & extraction: OpenAI API, structured outputs, Pydantic, document extraction, Playwright, prompt design and evaluation

Reliability & governance: Soda, pipeline monitoring, data quality, metadata, RBAC, row-level security, alerting

Cloud & delivery: Azure, AWS, Docker, Git, GitHub Actions, CI/CD, React, Flutter

EARLIER EXPERIENCE

One Acre Fund - Finance Systems Data Integration Intern | Mar 2017 - Aug 2017

Built ETL workflows and optimized warehouse structures for analytics.

PwC - Technology Advisory Intern, Boston, USA | 2015 - 2016

Supported technology consulting, systems-integration planning, KPI analysis, and data-migration engagements.

EDUCATION

Bachelor of Science (B.Sc.) in Mathematics & Computer Science | Brown University | 2013 - 2017